Sage Hiring Team
Sage
New York, NY

June 2, 2026

To the Sage Hiring Team,

I am writing to express my strong interest in the Senior Applied Machine Learning Engineer position on the Detect team. As an ML engineer who has spent the last eight years building and scaling predictive systems where accuracy has direct, real-world consequences, I am deeply drawn to Sage's mission. I specialize in owning the end-to-end machine learning lifecycle from data infrastructure through rigorous evaluation and deployment, and I am eager to help mature your early-stage models into robust, production-grade systems.

Throughout my career, I have gravitated toward high-stakes problems. While serving as a Senior Data Scientist and Technical Lead at the Texas Advanced Computing Center (TACC), I developed real-time flood inundation models and DARPA-funded disaster resiliency tools. In those environments, model reliability directly informed emergency response and life-saving interventions. The camera-based fall detection system at Sage presents the exact same class of responsibility, where improving model accuracy and reducing false positives directly impacts human lives and caregiver trust.

Building the infrastructure to guarantee that accuracy at scale is my core competency. Most recently, as the ML architect for Homecastr, I designed our end-to-end experimentation platform and evaluation infrastructure from scratch. Rather than simply training models in isolation, I built an automated evaluation suite that rigorously tests every new model iteration for probabilistic calibration, interval coverage, and accuracy before it is ever cleared for production. This framework allows for rapid iteration on new model architectures while mathematically guaranteeing that our deployed models remain highly reliable.

I am excited by the opportunity to bring this same level of rigorous evaluation discipline and MLOps expertise to Sage's multi-modal vision models. The challenge of scaling the Detect platform, including building repeatable fine-tuning pipelines and expanding detection capabilities beyond falls, is exactly the type of zero-to-one infrastructure transition where I excel.

As a New York City-based engineer who thrives in cross-functional, mission-driven teams, I am prepared to commute to the Union Square office and make an immediate impact on a product that seniors and caregivers rely on daily. Thank you for your time and consideration. I look forward to discussing how my background can help advance Sage's critical mission.

Sincerely,

Daniel Hardesty Lewis

Attached: resume

Daniel Hardesty Lewis
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Daniel Hardesty Lewis

Staff ML Engineer – production training, inference, evaluation, reliability

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infrastructure	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
Additional	CUDA; C++; TypeScript; React

Experience

Professional

Nov. 2024 – Present **Co-founder**, *Homecastr*, New York, NY
Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.
- Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
- Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
- Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
- Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z and MetaProp**.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA
AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation
- Specified an agentic workflow for BOM parsing and tariff simulation using Airflow, Elasticsearch embeddings, and FastAPI on Supabase and Redis.
- Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY
High-resolution elevation data and web delivery for engineering and hazard workflows.
- Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
- Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing $1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Math benchmarking and finetuning of Deepseek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki   Bikepacking US West & East Coasts

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Languages

Spanish Native Proficiency

French Professional Working Proficiency

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